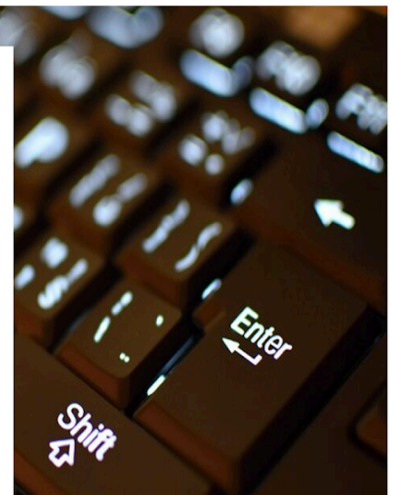


Computer Awareness

STUDY NOTES

Memory

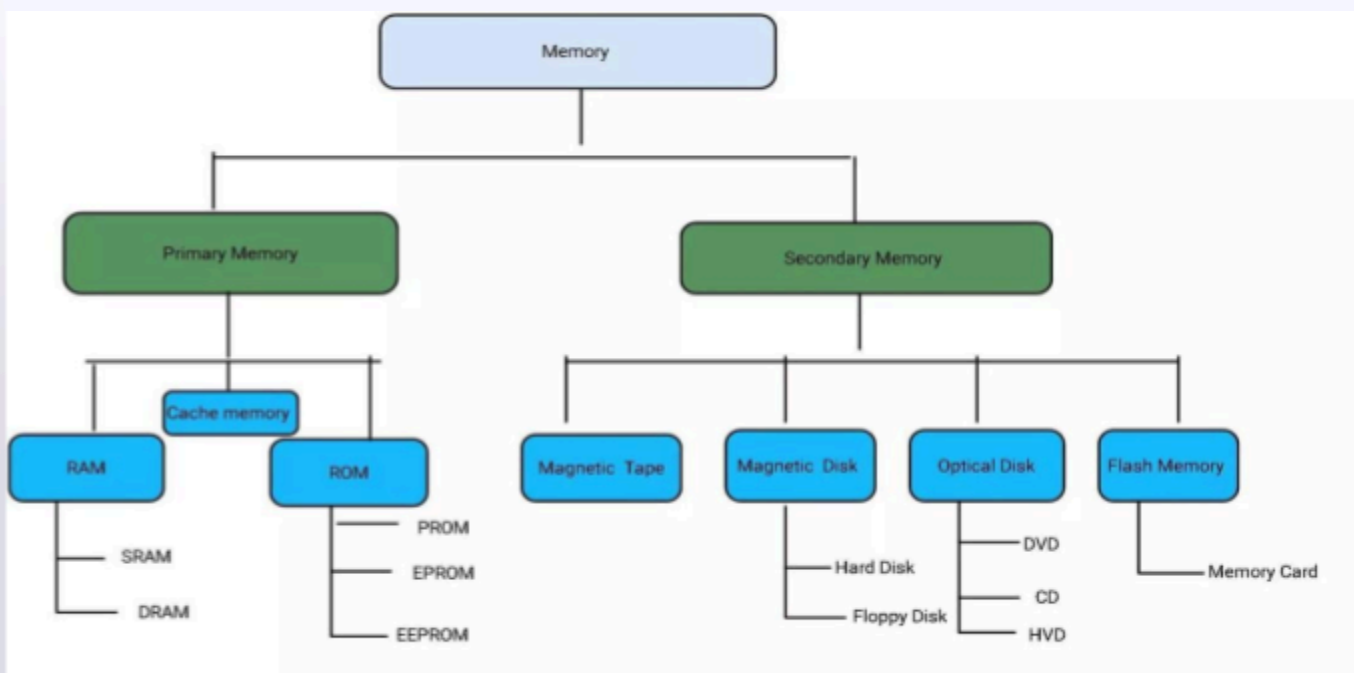


Memory

- A memory acts like a human brain. It is available to hold data and instructions. Computer memory is the type of storage space built in the computer, where data is to be processed and instructions needed for processing are stored.
- The memory is categorized into a high number of small sections called cells. Every memory location or cell consists of a unique address that varies from 0 to memory size -1. For instance, if the computer comprised of 64k words, then this memory entity has $64 * 1024 = 65536$ memory locations. The memory address of these locations differs from 0 to 65535.
- Memory is primarily of three types:
 - ◇ Cache Memory
 - ◇ Primary Memory/Main Memory
 - ◇ Secondary Memory

Cache Memory

Cache memory is the most **high speed semiconductor memory** type which can speed up the working of CPU. This works like a **buffer between the main memory and the CPU**. This memory is available to store those sections of data and programs which are very frequently required by the CPU. The sections of data and programs are transferred from the **disk to cache** memory through the operating system, from where the CPU has accessibility of them.



Advantages

- Cache memory is faster than main memory.
- It consumes less access time as compared to main memory.
- It stores the program that can be executed within a short period of time.
- It stores data for temporary use.

Disadvantages

- Cache memory has limited capacity.
- It is very expensive.

Primary Memory

- Primary memory keeps only those data and instructions on which the computer is currently processing. It has a fixed capacity and data is lost when power is disconnected. It is usually made up of semiconductor ICs and registers. These memories are not as fast as registers. The fetched data and instruction going to be processed resides in the main memory. It is categorized in 2 subcategories RAM and ROM.

- **Characteristics of Main Memory:**

- ◇ These are semiconductor memories.
- ◇ It is known as the main memory.
- ◇ Usually volatile memory.
- ◇ Data is lost in case power is switched off.
- ◇ It is the working memory of the computer.
- ◇ Faster than secondary memories.
- ◇ A computer cannot run without the primary memory.

Secondary Memory

- This kind of memory is also called external memory or non-volatile memory. It works slower as compared to the main memory. These are installed for holding data/ information permanently. CPU does not access these memories directly, rather they are accessed through input-output routines. The data hold in secondary memories are transferred to the main memory for, and then the CPU can access that data. For instance, disk, CD-ROM, DVD, etc.

- **Characteristics of Secondary Memory:**

- ◇ These are magnetic and optical memories.
- ◇ It is known as the backup memory.
- ◇ It is a non-volatile memory.
- ◇ Data is permanently stored even if power is switched off.
- ◇ It is used for storage of data in a computer.
- ◇ Computers may run without the secondary memory.
- ◇ Slower than primary memories.

Random Access Memory

- RAM (Random Access Memory) is the kind of internal memory of the CPU for **holding data**, program, and **program result after processing**. This memory is read/write memory which holds data **until the machine is supplied with power**. As soon as the machine is powered off, data is dumped.
- **Access time in RAM** is not dependent on the address, that is, each storage location residing in the memory is as simple to be accessed as other locations and takes the same duration of time. Data in the RAM can be traversed randomly but it is comparatively very expensive.
- RAM is a volatile type of memory, i.e. data stored in it is dumped when we power off the computer or if there is a power failure situation. Hence, it is recommended to have an Uninterruptible Power System (UPS) which can provide enough backup power to store data before shut down. The RAM is small in terms of its physical size as well as in terms of the amount of data it can store.
- RAM is of two types:
 - ◇ Static RAM (SRAM)
 - ◇ Dynamic RAM (DRAM)

Static RAM (SRAM)

- The word static implies that the memory holds its contents as long as the electricity is being supplied. But data is dumped when the power gets down because of its volatile nature. SRAM chips are equipped with a matrix of 6-transistors and no capacitors. Transistors are not dependent on power to prevent leakage, hence SRAM need not be refreshed on a regular interval.
- In the matrix, there exists an extra space, that is why SRAM consists of a high

number of chips as compared to the DRAM for the same capacity of storage space, making the manufacturing costs comparatively higher. That is why SRAM is utilized as cache memory and has very fast accessing capability.

- **Characteristic of Static RAM:**

- ◇ Long life
- ◇ No need to refresh
- ◇ Faster
- ◇ Used as cache memory
- ◇ Large size
- ◇ Expensive
- ◇ High power consumption

Dynamic RAM (DRAM)

- Unlike SRAM, DRAM must be continually refreshed so that the data should be maintained. This is achieved by placing the memory on a refresh circuit which rewrites the content several hundred times every second.
- DRAM is installed for most system memory as it is comparatively cheap and small. All DRAMs consist of memory cells, which are composed of a single capacitor and single transistor.

- **Characteristics of Dynamic RAM:**

- ◇ Short data lifetime
- ◇ Needs to be refreshed continuously
- ◇ Slower as compared to SRAM
- ◇ Used as RAM
- ◇ Smaller in size
- ◇ Less expensive
- ◇ Less power consumption

Read Only Memory

- **ROM** is an acronym for Read Only Memory. The memory from which the data can be read only but **rewriting it is not possible**. This kind of memory is **non-volatile** in nature. The data is stored **permanently** in such memories at the time of manufacturing. A ROM holds such instructions that are needed to turn on a computer. This process is called bootstrap. ROM chips are **not only installed in the computer** but also in **other electronic devices** like washing machines and microwave ovens.

Masked ROM (MROM)

- The early ROMs were **hard-wired instruments** that consisted of a pre-programmed set of **data or instructions**.
- These types of ROMs are called masked **ROMs**, which are comparatively **inexpensive**.

Programmable Read Only Memory (PROM)

- PROM is a type of **read-only memory** that can be **amended only once by a user**.
- The user buys an empty PROM and input the required data using a **PROM program**. It consists of the **small fuses** inside which are **burnt open** during programming.
- It is possible to program this memory **only once** and is **not erasable**.

Erasable & Programmable Read Only Memory (EPROM)

- It is possible to erase EPROM by exposing it to ultra-violet light for a period of up to 40 minutes. Normally, an EPROM eraser performs this operation. An electrical charge is stored in an insulated gate region while programming.
- The charge is held for more than 10 years as there is no leakage path. In order to erase this charge, ultra-violet light is passed via a quartz crystal window (lid).
- This process of exposure to UV light dissipates the charge. The quartz lid is hidden with a sticker during normal usage.

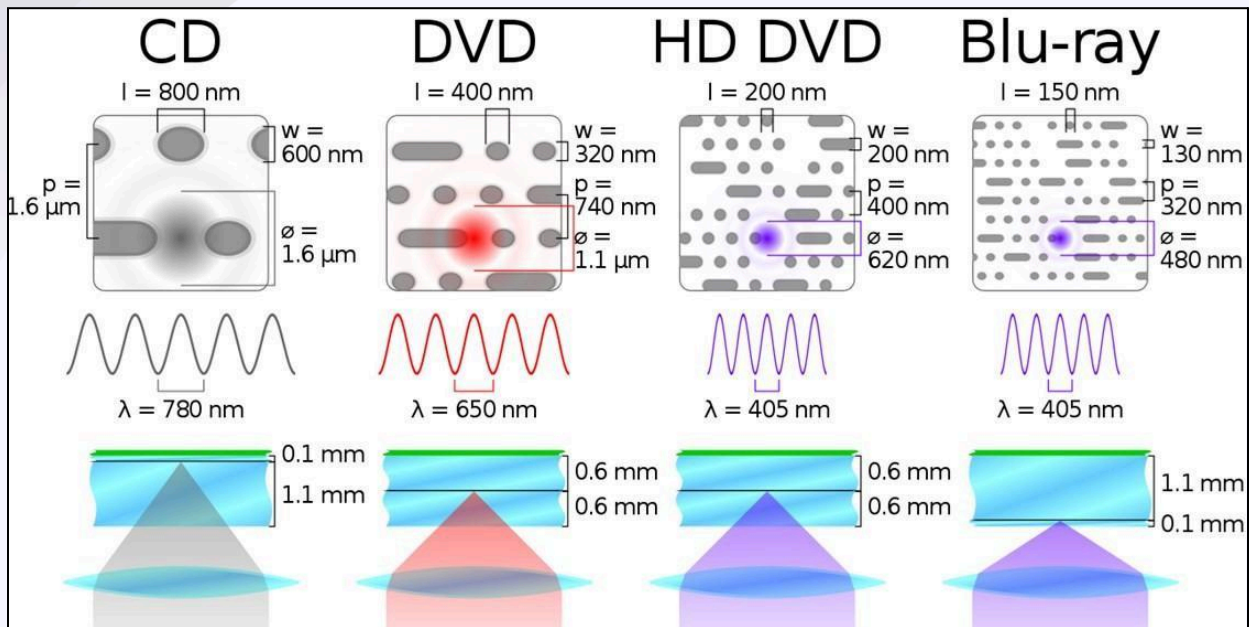
Electrically Erasable & Programmable Read Only Memory (EEPROM)

- EEPROM can be programmed and erased using electricity. It is possible to erase and reprogram it about ten thousand times.
- Erasing or programming, both take about 4 to 10 ms (millisecond) time. In EEPROM, any desired location can be separately erased and programmed.
- It is possible to erase EEPROMs at the rate of one byte at a time, instead of erasing the entire chip all at once. That is why the technique of reprogramming is quite flexible but slow.

Advantages of ROM

- Non-volatile in nature

- Cannot be accidentally changed
- Cheaper than RAMs
- Easy to test
- More reliable than RAMs
- Static and do not require refreshing
- Contents are always known and can be verified



Computer Memory Unit

- A memory unit is the amount of data that can be stored in the storage unit. This storage capacity is expressed in terms of Bytes.
- The following table explains the main memory storage units -

S.No.	Unit & Description
1	Bit (Binary Digit) A binary digit is logical 0 and 1 representing a passive or an active state of a component in an electric circuit.
2	Nibble A group of 4 bits is called nibble.

3	Byte A group of 8 bits is called byte. A byte is the smallest unit, which can represent a data item or a character.
4	Word A computer word, like a byte, is a group of fixed number of bits processed as a unit, which varies from computer to computer but is fixed for each computer. The length of a computer word is called word-size or word length. It may be as small as 8 bits or may be as long as 96 bits. A computer stores the information in the form of computer words.

5	Kilobyte Kilobyte can be defined as one of the many multiples of the byte memory units. The prefix kilo- , normally, represents the unit to be 'Kilo' and it comes in multiples of '1000'. Though in terms of international standards kilo means 1000, in a typical memory unit, the binary digits are seen as Base 2 and so the 'Kilo' here means 1024 of binary digits or bits. In short, 1Kilobyte or 1KB is equal to the 1024 bits or binary digits.
6	Megabyte Similar to the Kilobyte memory unit, the megabyte memory unit is used when the bits are large in number and when it makes it easy to say the number to be in lesser length. That is, in international standards the prefix 'mega-' is used for showing the 10^6 or the 1000000. This also means that a typical MB is shown with Base 2, and so the 1MB or the 1 Megabyte equals the 1048576 bytes of the memory unit. It can also be calculated as $(1024)^2$ bytes or the 2^{20} bytes.
7	Gigabyte The Gigabytes are used to represent the data to be in multiples of 10^9 or 1000000000 binary digits, and when the memory unit is in terms of Base 2 then the storage space is calculated as 2^{30} bytes or as 1024^3 bytes. In other words, the memory units used by the processor that runs on the Base 2 are termed the Giga- bytes and are equal to the 1073741824 bytes.

8	Terabyte Terabytes are applied when the memory requirement is as high as 2^{40} , that is the binary digits exponential to the number 40. It can also be seen as an equal to the 1024 terabytes and is represented as TB. Or in other words equals 10^{12} bytes in international standards and that can also be termed as the trillion bytes.
9.	Petabyte Petabyte can be defined as a memory unit that represents the higher multiples of bytes, next to the Terabyte that comes in PB units. It is equal to the 10^{15} bytes in terms of the international standards, or the 1024^5 bytes when the bytes come with the Base 2.
10	Exabyte Exabyte is a memory unit that is equal to the 10^{18} equivalent in the international standards scale, and the same is equal to the 1024^6 for the memory units that come in the form of Base 2. The unit terms of the Exabyte are EB, and this EB is the same as the 1 million TB or the 1 billion GB or the 1 thousand PB.
11	Zettabyte Similar to other byte sized memory units, Zettabyte can be seen as an equivalent to 10^{21} , according to the 'zetta' representation in terms of the international standards. The Zettabyte or the ZB also can be seen as 1 thousand of the EB, 1 million PB, 1 billion TB, 1 trillion GB, etc.
12	Yottabyte As the name says, the Yottabyte unit or the YB represents another memory unit for the bytes to be in multiples of the other lower byte units or even the bytes. In international standards, it is equal to the 10^{24} , and for base 2 byte units the YB is also equal to 1024^8 , 1 thousand ZB, 1 trillion TB, etc.

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Units of Computer Memory Measurements

1 Bit	= Binary Digit
8 Bits	= 1 Byte
1024 Bytes	= 1 KB [Kilo Byte]
1024 KB	= 1 MB [Mega Byte]
1024 MB	= 1 GB [Giga Byte]
1024 GB	= 1 TB [Terra Byte]
1024 TB	= 1 PB [Peta Byte]
1024 PB	= 1 EB [Exa Byte]
1024 EB	= 1 ZB [Zetta Byte]
1024 ZB	= 1 YB [Yotta Byte]
1024 YB	= 1 Bronto Byte
1024 Brontobyte	= 1 Geop Byte

Geop Byte is the Highest Memory.